

Symmetric Group.

Def. Let A be a non-empty set.

A Group of bijections on A :

$$S_A \stackrel{\text{def}}{=} \{ f: A \rightarrow A \mid f \text{ is bijection} \}$$

is called a Symmetric Group on A .

Lemma. Define S_n as Symmetric group on a set $\{1, 2, \dots, n\}$.

If $|A| = n$, then $S_A \cong S_n$.

Proof. Since $|A| = n$, there exists a bijection

$$f: A \rightarrow \{1, 2, \dots, n\}$$

Define a map:

$$\mathcal{Q}: S_A \rightarrow S_n; \sigma \mapsto f \circ \sigma \circ f^{-1}$$

$A \rightarrow A \quad A \rightarrow \{1, \dots, n\} \quad \{1, \dots, n\} \rightarrow A$

It is well defined, because Composition of bijection is bijection.

And, injectiveness is given by just Cancellation;

$$f \circ \sigma_1 \circ f^{-1} = f \circ \sigma_2 \circ f^{-1} \Rightarrow \sigma_1 = \sigma_2$$

Surjectivity is given by: for any $\alpha \in S_n$, $f^{-1} \circ \alpha \circ f$ is contained in S_A and

$$\mathcal{Q}(f^{-1} \circ \alpha \circ f) = f \circ f^{-1} \circ \alpha \circ f \circ f^{-1} = \alpha.$$

Finally, Homomorphism because: for any $\sigma_1, \sigma_2 \in S_A$,

$$\mathcal{Q}(\sigma_1 \circ \sigma_2) = f \circ \sigma_1 \circ \sigma_2 \circ f^{-1} = f \circ \sigma_1 \circ f^{-1} \circ f \circ \sigma_2 \circ f^{-1} = \mathcal{Q}(\sigma_1) \circ \mathcal{Q}(\sigma_2).$$

Now, we denote all permutation group as S_n .

Cycle Decomposition.

Let S_n be a Symmetric Group. Then, every element of S_n can be represented by

$$\sigma = \begin{pmatrix} 1 & 2 & \dots & n \\ a_1 & a_2 & \dots & a_n \end{pmatrix} \text{ where } \{a_i\}_{i=1}^n \text{ is a sequence having range as } \{1, \dots, n\}.$$

Choose $a \in \{1, \dots, n\}$, Construct a Cycle: for the smallest positive integer $k_1 \geq 1$ s.t. $\sigma^{k_1}(a) = a$,

$$(a \ \sigma^1(a) \ \sigma^2(a) \ \dots \ \sigma^{k_1-1}(a))$$

Choose $b \in \{1, \dots, n\}$ s.t. $\forall 0 \leq i \leq k_1-1, b \neq \sigma^i(a)$. Then, similarly, we can construct:

$$(b \ \sigma^1(b) \ \sigma^2(b) \ \dots \ \sigma^{k_2-1}(b))$$

Since $\{1, \dots, n\}$ is finite set, this process will terminate in finite time.

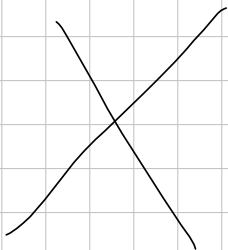
And, the elements in a cycle are mutually distinct, because: if for some $0 \leq i < j \leq k_1-1$,

$$\sigma^i(a) = \sigma^j(a) \Rightarrow \sigma^0(a) = a = \sigma^{j-i}(a)$$

But this contradicts to minimality of k_1 .

Moreover, these two cycles are disjoint: if for some $0 \leq r \leq k_1-1, 0 \leq t \leq k_2-1$,

$$\sigma^r(a) = \sigma^t(b) \Rightarrow b = \sigma^{r-t}(a)$$



Cycle decomposition.

Let S_A be a symmetric group of a set A . Let $\sigma \in S_A$ be given.

Define a relation on A as:

$$a \sim b \iff a = \sigma^i(b) \text{ for some } i \in \mathbb{Z}.$$

Then, it is an equivalent relation because

$a \sim a$ is trivial, put $i=0$.

If $a \sim b$, then $a = \sigma^i(b)$ for some $i \in \mathbb{Z}$. Take σ^{-i} to both side, then $\sigma^{-i}(a) = b$.

If $a \sim b$ and $b \sim c$, then, for some $i, j \in \mathbb{Z}$, $a = \sigma^i(b)$ and $b = \sigma^j(c)$. Now,
$$a = \sigma^i(b) = \sigma^i(\sigma^j(c)) = \sigma^{i+j}(c)$$

Therefore, we can consider a partition

$$A/\sim = \{C_a \mid a \in A\} \text{ where } C_a = \{\sigma^i(a) \mid i \in \mathbb{Z}\}.$$

If A is infinite, A can have infinitely many infinite orbits.

But, if A is finite, A must have finitely many finite orbits.

Regardless of Cardinality of A , the orbits have the following properties:

Lemma. Let A be a set, and let $\sigma \in S_A$ be a permutation on A .

If $a \in A$ satisfies that $\sigma^n(a) = a$ for some $n \geq 0$,

then, for $p = \min \{n \geq 0 \mid \sigma^n(a) = a\}$,

$$\{a, \sigma(a), \dots, \sigma^{p-1}(a)\} = C_a$$

and each $\sigma^i(a)$ ($0 \leq i \leq p-1$) are mutually disjoint.

Proof. The existence of such p is given by the Well-ordering principle.

And, since $\sigma^p(a) = a$, for any $k \in \mathbb{Z}$,

$$\sigma^k(a) = \sigma^{k \bmod p}(a).$$

Finally, to show that mutual disjointness, suppose that for some $0 \leq i < j \leq p-1$,

$$\sigma^i(a) = \sigma^j(a)$$

Take σ^{-i} to both side, then

$$a = \sigma^{j-i}(a)$$

But it is contradiction to the minimality. $\square$

If exists, the smallest number $p \geq 1$ s.t. $\sigma^p(a) = a$ is called period of a with respect to σ .

Clearly, if C_a is finite set, then all elements have a same period.

Def. Let $\sigma \in S_n$ be a permutation

If σ has at most one cycle having more than one element,
then σ is called a Cycle, and denote

$$\sigma = (a \ \sigma(a) \ \sigma^2(a) \ \dots \ \sigma^{p-1}(a)) \text{ where } p \geq 1 \text{ is period of } a.$$

Thm. Let S_n be a Symmetric Group on n letters.

Every permutation of S_n can be represented by finite composition of disjoint cycles.
Moreover, disjoint cycles commute.

Proof. Let $\sigma \in S_n$ be given. Then, there exist finite orbits of σ :

$$C_{a_1}, \dots, C_{a_k}.$$

For each $1 \leq i \leq k$, define:

$$\sigma_i : \{1, \dots, n\} \rightarrow \{1, \dots, n\} : x \mapsto \begin{cases} \sigma(x) & \text{if } x \in C_{a_i} \\ x & \text{if } x \notin C_{a_i} \end{cases}$$

Then, clearly each σ_i 's are Cycles, and commute. Moreover,

$$\sigma = \sigma_1 \circ \dots \circ \sigma_k.$$

Transposition decomposition

Def. Let S_n be a Symmetric Group on n letters.

A length of 2 cycle is called a transposition in S_n .

Thm. Every Cycles in S_n Can be represented by finite product of transpositions.

Proof. Let a Cycle $(a_1 a_2 \dots a_k) \in S_n$ be given. Then, by computation,

$$(a_1 a_2 \dots a_k) = (a_1 a_k)(a_1 a_{k-1}) \dots (a_1 a_2)$$

Now, we can consider all permutation in S_n is a product of transpositions.

Note: transposition decomposition is not unique. for example,

$$(1 2 3) = (1 3)(1 2) = (1 3)(1 2)(1 3)(1 3)$$

Thm. No permutation in S_n can be represented by both even and odd product of transpositions.

Proof. Let $\sigma \in S_n$ be given. To use contradiction, Suppose that

$$\sigma = f_1 \circ \dots \circ f_{n_1} = g_1 \circ \dots \circ g_{n_2}$$

where f_i, g_i are transposition, and n_1 is odd, n_2 is even number.

Consider the standard basis for $\mathbb{R}^n$,

$$\{e_1, \dots, e_n\}$$

clearly,

$$\det(e_1 \dots e_n) = 1.$$

$$\text{And, } \det(e_{f_1(1)} e_{f_1(2)} \dots e_{f_1(n)}) = -1$$

Because it is the change of two rows for I_n .

$$\text{Inductively, } \det(e_{f_1 \circ \dots \circ f_{n_1}(1)} \dots e_{f_1 \circ \dots \circ f_{n_1}(n)}) = (-1)^{n_1} = -1$$

$$\text{and } \det(e_{g_1 \circ \dots \circ g_{n_2}(1)} \dots e_{g_1 \circ \dots \circ g_{n_2}(n)}) = (-1)^{n_2} = 1.$$

Contradiction.

Thm. Let S_n be a Symmetric Group on n letters.

1. Let $\sigma \in S_n$ is a cycle of length m . Then

σ^i is m -cycle if and only if $\gcd(i, m) = 1$

Proof. Denote $\sigma = (a_0 a_1 \dots a_{m-1})$ as cycle notation. That is, $\sigma^k(a_i) = a_{i+k \bmod m}$.

Let $2 \leq i \leq m-1$ be given.

σ^i is m -cycle $\Leftrightarrow a_i, \sigma^i(a_i), \sigma^{2i}(a_i), \dots, \sigma^{(m-1)i}(a_i)$ are distinct

$\Leftrightarrow a_i, a_{2i \bmod m}, a_{3i \bmod m}, \dots, a_{mi \bmod m}$ are distinct.

$\Leftrightarrow \forall x \neq y \in \{1, \dots, m\}, x \neq y \Rightarrow xi \not\equiv yi \pmod m$

$\Leftrightarrow \gcd(i, m) = 1$.

2. $\sigma \in S_n$ has order 2 iff σ is product of commute transpositions.

Proof. Let $\sigma \in S_n$ has order 2. That is, for all $1 \leq i \leq n$, $\sigma^2(i) = i$ and $i \neq \sigma(i)$.

Now, a cycle in the cycle must be identity or transposition.

Since order of σ is not 1, it contains at least one transposition.

The converse is trivial.

3. Let p be a prime number. Then,

$\sigma \in S_n$ has order p iff σ is product of commute p -cycles.

4. Let $\sigma \in S_n$ be a permutation with cycle decomposition

$$\sigma = \sigma_1 \circ \dots \circ \sigma_k.$$

Then,

$$|\sigma| = \text{lcm}(|\sigma_1|, \dots, |\sigma_k|).$$

Note: we already know that the map

$$\text{sgn}: S_n \rightarrow \{\pm 1\}: \sigma \mapsto \begin{cases} 1 & \text{if } \sigma \text{ is even} \\ -1 & \text{if } \sigma \text{ is odd} \end{cases}$$

is well-defined.

Lemma. $\text{sgn}: S_n \rightarrow \{\pm 1\}$ is Epimorphism.

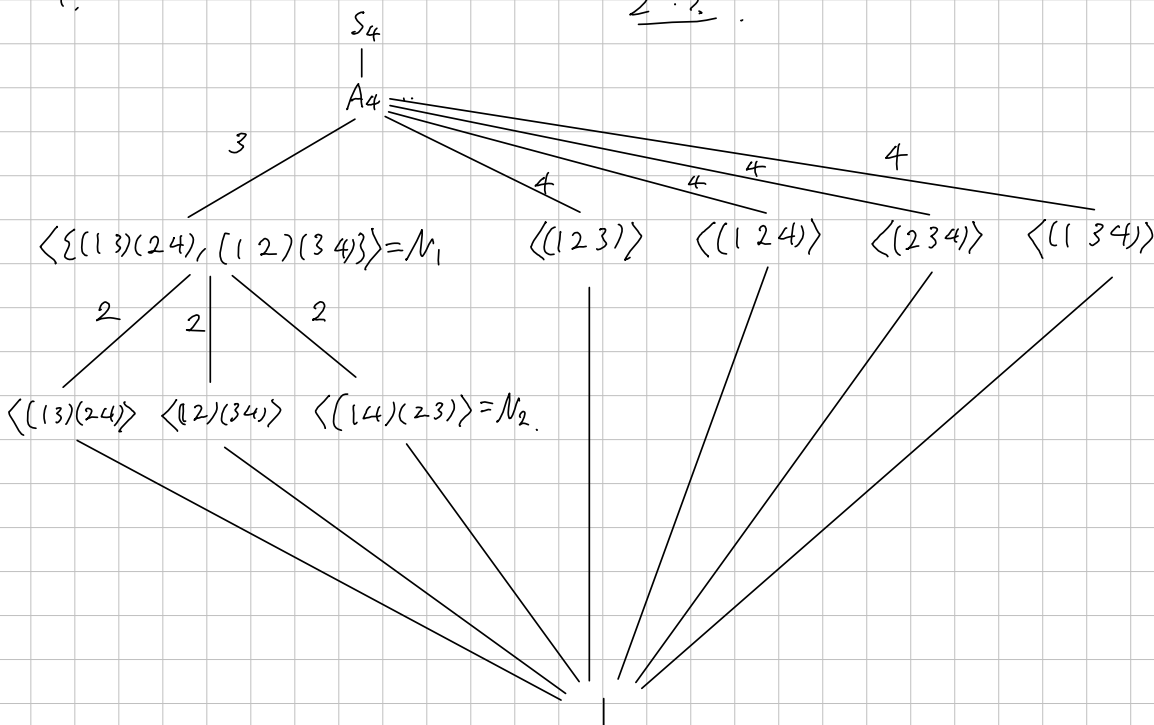
Def. Let S_n be a Symmetric Group on n letters.

A subgroup $A_n \stackrel{\text{def}}{=} \ker(\text{sgn})$ is called Alternating Group of degree n .

Clearly, $A_n \trianglelefteq S_n$ and $|S_n:A_n| = 2$, because $S_n/A_n \cong \{\pm 1\}$.

S_4 ,

$$2^2 \cdot 3$$



Thm. S_4 is Solvable.

Proof. Since N_1 is subgroup of all elements of order 2, and

Conjugate Conjugate change the order, $N_1 \trianglelefteq A_4$.

Now, $1 \trianglelefteq N_2 \trianglelefteq N_1 \trianglelefteq A_4 \trianglelefteq S_4$ all factors above are $\mathbb{Z}_2$ or $\mathbb{Z}_3$, thus abelian.

Thm. Conjugation in S_n does not change the cycle structure.

Thm. A_5 is